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ID Number 2: _____

(This page will not be seen by the grader.)

Academic Integrity Statement: I have not used AI or other internet resources to assist with these problems.

Partner 1 signature: _____

Partner 2 signature: _____

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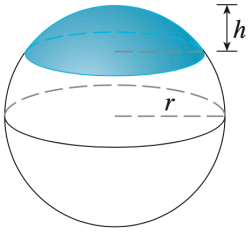
Instructions: Solve each of the following problems, showing your work.

1. Sketch the region enclosed by the curves $4x + y^2 = 12$ and $x = y$, and find the area of the region.

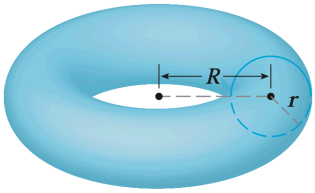
2. A hole of radius r is bored through the center of a sphere of radius $R > r$. Find the volume of the remaining portion of the sphere.

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3. Find the volume of a cap of a sphere of radius r and height h.



4. Use the Method of Cylindrical Shells to find the volume of a solid torus with radii r and R.



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5. Use trig substitution to find the exact length of the parabola $y = x^2$ from $(0,0)$ to $(2,4)$.